

Cerberus - How Radix achieves infinite linear scalability while preserving atomic composability

Conclusion

16

Before we wrap everything up, we go back to the very beginning:

Cerberus - How Radix achieves infinite linear scalability while preserving atomic composability

To summarize, Cerberus' novel "braided" consensus design is what provides Radix the unique ability to compose transactions and settle them atomically across shards.

Cross-shard atomic composability, in turn, lets Cerberus take full advantage of partial ordering (as provided by the Radix Engine):

- If substates are unrelated, they can be processed in parallel across the vastness of the shardspace. The practically infinite shardspace allows Radix to process more transaction throughput just by adding more nodes - there will always be room for more!
- If substates are related, Radix Engine tells Cerberus they must be processed atomically in a single cross-shard round of consensus - thus preserving atomic composability.

So, Radix achieves practically infinite linear scalability *precisely because* cross-shard atomic composability is baked into the very fabric of every transaction. Atomic composability is not an afterthought to scalability - it is the key enabler!

And so it all comes back to braiding of consensus. Of all our technical advancements, *this* is Radix's most profound innovation. Braiding is the root of Radix's new paradigm in DLT design.

Unique Features



Practically Infinite Linear Scalability

Allows Radix to always scale to meet the needs of network usage – just like the internet, there is no ceiling



Cross-Shard Atomic Composability

Allows transactions to be composed across shards - either settling all together, or not at all - across the global ledger

In final conclusion, we revisit the Blockchain Dilemma:

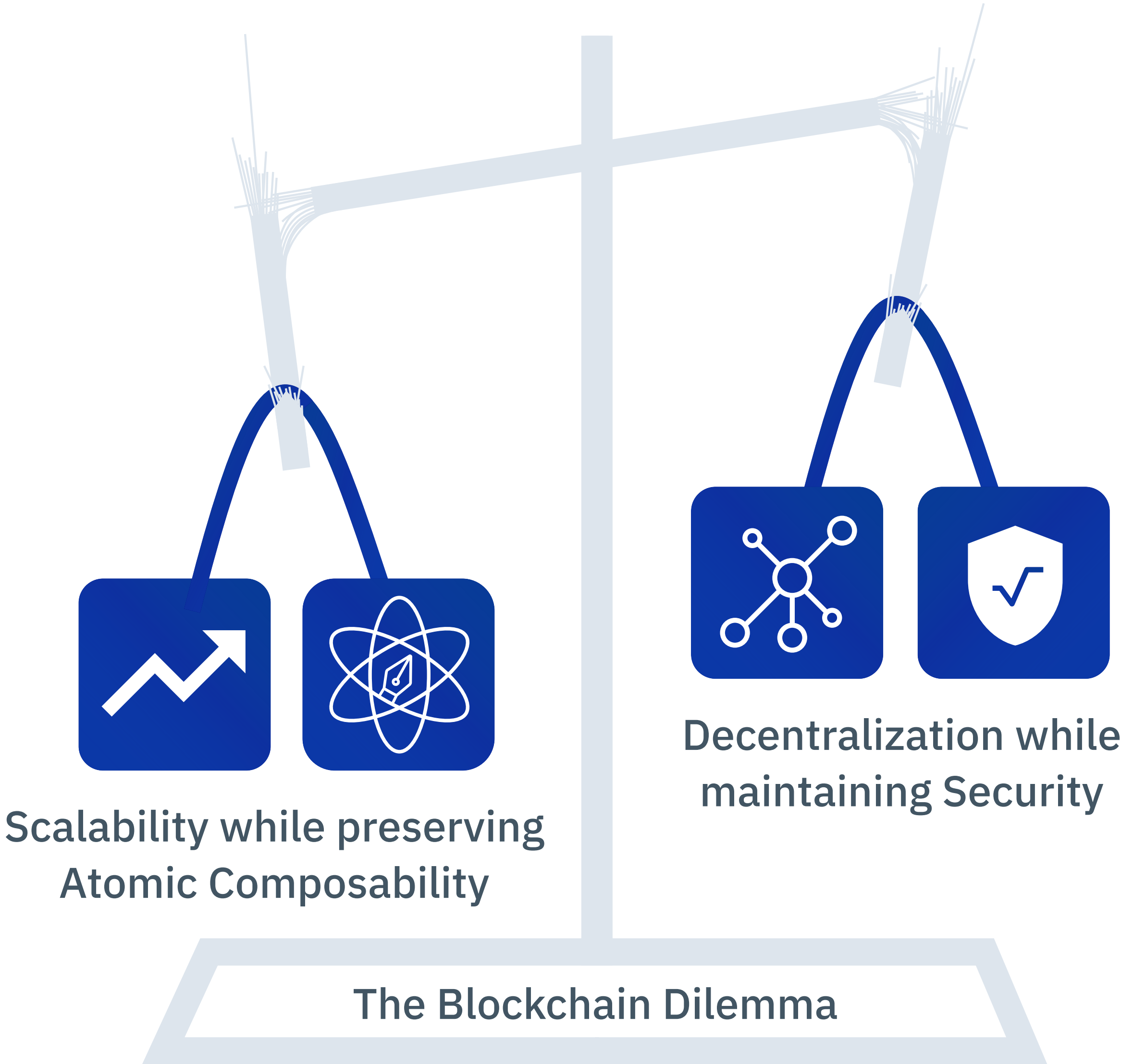
Scalability while preserving atomic composability

Radix offers practically infinite scalability while preserving atomic composability. No Layer 2 solutions will ever be necessary. Settlement finality can be achieved within 5 seconds because there is no need to wait to see if a longer chain emerges.

Decentralization while maintaining Security

Radix has no masternodes or coordinators; there are no relay, beacon, or runner chains. Radix exists as a single layer of programmable trust as all nodes are created equal. As there is no limit to how many nodes can validate on Radix, and that nodes can always be run on simple hardware, Radix is as decentralized as can be.

For security, Cerberus' unique braided consensus design has been [publicly reviewed by UC Davis' ExpoLab consensus experts](#) and others and shown to provide strong guarantees of safety and liveness. Radix uses Delegated Proof of Stake for Sybil protection and has the same resistance to Sybil attacks as any other DPoS protocol.



As a final thought, Radix rewrites the standard for what was previously thought impossible. Radix demonstrates that it is possible to achieve the scalability and atomic composability required to scale DeFi to billions of people *without* sacrificing decentralization and security.

The version of Cerberus described in this infographic series is scheduled to launch as part of the fully sharded Radix Xi'an release. Please visit www.radixdlt.com for details on the Radix roadmap.